Practice Set B

Class 12 Mathematics, Chapter 10: Vector Algebra

Maximum marks: **47** **20** questions

Name _____ Date _____ Score _____ / 47

1. All questions are compulsory. Marks are shown against each question.
2. Show every step of working: steps carry marks, not just the final answer.
3. Where a question offers OR, attempt only one of the two choices.

Q No.	Question	Marks

Level 1

8 marks

1 MCQ, Board-pattern — 1 mark

The unit vector $\hat{n}$ perpendicular to both $\vec{a} = \hat{i} + \hat{j}$ and $\vec{b} = \hat{j} + \hat{k}$, such that $\vec{a}$, $\vec{b}$, $\hat{n}$ form a right-handed system, is:

(A) $\dfrac{-\hat{i} + \hat{j} - \hat{k}}{\sqrt{3}}$ **(B)** $\dfrac{\hat{i} + \hat{j} + \hat{k}}{\sqrt{3}}$ **(C)** $\hat{i} - \hat{j} + \hat{k}$ **(D)** $\dfrac{\hat{i} - \hat{j} + \hat{k}}{\sqrt{3}}$

2 MCQ, NCERT Exemplar — 1 mark

If $\vec{a}$, $\vec{b}$, $\vec{c}$ are three vectors such that $\vec{a} + \vec{b} + \vec{c} = \vec{0}$ and $|\vec{a}| = 2$, $|\vec{b}| = 3$, $|\vec{c}| = 5$, then the value of $\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}$ is:

(A) 0 **(B)** 1 **(C)** -19 **(D)** 38

3 MCQ, Board-pattern — 1 mark

If $\vec{a} = \hat{i} + \lambda\hat{j} + 2\hat{k}$ and $\vec{b} = 2\hat{i} - \hat{j} + \hat{k}$, then the value(s) of λ for which $\vec{a} + \vec{b}$ is perpendicular to $\vec{a} - \vec{b}$ is/are:

(A) 1 only **(B)** ± 1 **(C)** 4 **(D)** 0

4 MCQ, Board-pattern — 1 mark

If $|\vec{a}| = 2$, $|\vec{b}| = 3$ and $|\vec{a} + \vec{b}| = \sqrt{19}$, then $|\vec{a} \times \vec{b}|$ is equal to:

(A) $3\sqrt{3}$ **(B)** 3 **(C)** 27 **(D)** 6

5 MCQ, NCERT Exemplar — 1 mark

If A, B, C, D are the points with position vectors $\hat{i} + \hat{j} - \hat{k}$, $2\hat{i} - \hat{j} + 3\hat{k}$, $2\hat{i} - 3\hat{k}$, $3\hat{i} - 2\hat{j} + \hat{k}$ respectively, then the projection of $\overrightarrow{AB}$ along $\overrightarrow{CD}$ is:

(A) $\dfrac{1}{\sqrt{21}}$ **(B)** 21 **(C)** $\sqrt{21}$ **(D)** $-\sqrt{21}$

Q No.	Question	Marks
6	Assertion-Reason, Board-pattern **Assertion (A):** The vectors $\vec{a} = 2\hat{i} - 3\hat{j} + 4\hat{k}$ and $\vec{b} = -4\hat{i} + 6\hat{j} - 8\hat{k}$ are collinear. **Reason (R):** Two collinear vectors always have the same direction. **(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A). **(B)** Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A). **(C)** Assertion (A) is true, but Reason (R) is false. **(D)** Assertion (A) is false, but Reason (R) is true.	1 mark
7	Short answer, NCERT Exemplar Find the area of the parallelogram whose diagonals are $2\hat{i} - \hat{j} + \hat{k}$ and $\hat{i} + 3\hat{j} - \hat{k}$.	2 marks

Level 2

16 marks

8	Short answer, NCERT Exemplar A vector $\vec{r}$ has magnitude 14 and direction ratios $2, 3, -6$. Find the direction cosines and components of $\vec{r}$, given that $\vec{r}$ makes an acute angle with the x-axis.	2 marks
9	Short answer, NCERT Exemplar Find a vector of magnitude 6, which is perpendicular to both the vectors $2\hat{i} - \hat{j} + 2\hat{k}$ and $4\hat{i} - \hat{j} + 3\hat{k}$.	2 marks
10	Short answer, NCERT Exemplar The vectors $\vec{a} = 3\hat{i} - 2\hat{j} + 2\hat{k}$ and $\vec{b} = -\hat{i} - 2\hat{k}$ are the adjacent sides of a parallelogram. Find the acute angle between its diagonals.	2 marks
11	Short answer, Board-pattern Using vectors, find the area of the triangle with vertices $A(1, 0, 2)$, $B(2, 4, 1)$ and $C(3, 2, 3)$. Hence find the length of the perpendicular from A to BC.	3 marks
12	Short answer, PYQ-style Find λ if the projection of $\vec{a} = \lambda\hat{i} + \hat{j} + 4\hat{k}$ on $\vec{b} = 2\hat{i} + 6\hat{j} + 3\hat{k}$ is 4 units. Hence find the projection vector of $\vec{a}$ on $\vec{b}$.	3 marks
13	Short answer, Board-pattern If $\lvert\vec{a}\rvert = 2$, $\lvert\vec{b}\rvert = 1$ and the angle between $\vec{a}$ and $\vec{b}$ is $\dfrac{\pi}{3}$, find the value of λ for which $\vec{a} + \lambda\vec{b}$ is perpendicular to $\vec{a} - 3\vec{b}$.	2 marks
14	Short answer, NCERT Exemplar If $\vec{a}$ and $\vec{b}$ are the position vectors of A and B respectively, find the position vector of a point C in BA produced such that $BC = 1.5\,BA$.	2 marks

Level 3

23 marks

15	Short answer, NCERT Exemplar If $\vec{a} + \vec{b} + \vec{c} = \vec{0}$, show that $\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a}$. Interpret the result geometrically.	3 marks

Q No.	Question	Marks

16 Long answer, NCERT Exemplar — 5 marks

If $\vec{a}$, $\vec{b}$, $\vec{c}$ determine the vertices of a triangle, show that $\frac{1}{2}\left[\vec{b}\times\vec{c}+\vec{c}\times\vec{a}+\vec{a}\times\vec{b}\right]$ gives the vector area of the triangle. Hence deduce the condition that the three points $\vec{a}$, $\vec{b}$, $\vec{c}$ are collinear. Also find the unit vector normal to the plane of the triangle.

17 Short answer, Board-pattern — 3 marks

Find a vector of magnitude 9 units which is perpendicular to both $\vec{a}=2\hat{i}+\hat{j}+2\hat{k}$ and $\vec{b}=3\hat{i}+2\hat{j}+2\hat{k}$, and which makes an obtuse angle with the y-axis.

18 Short answer, Board-pattern — 3 marks

$ABCD$ is a plane quadrilateral with vertices $A(1,1,1)$, $B(3,1,3)$, $C(3,4,6)$ and $D(1,2,2)$. Using vectors, find its area by dividing it along the diagonal AC.

19 Long answer, Board-pattern — 5 marks

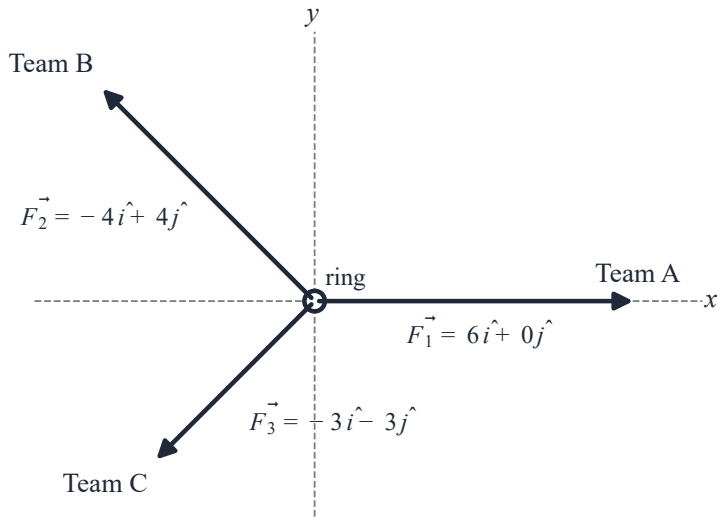
In a triangle ABC, let a, b, c be the lengths of the sides opposite to the vertices A, B, C respectively.

(i) Using vectors, prove that $\cos A=\dfrac{b^2+c^2-a^2}{2bc}$. — 3 marks

(ii) Hence find $\cos A$ for the triangle with vertices $A(1,2,1)$, $B(3,4,2)$ and $C(5,2,-2)$, and check your answer using $\overrightarrow{AB}\cdot\overrightarrow{AC}$. — 2 marks

20 Case study, CBSE SQP 2023-24 — 4 marks

Case Study: Teams A, B, C went for playing a tug of war game. Teams A, B, C have attached a rope to a metal ring and each is trying to pull the ring into its own area. Team A pulls with force $\vec{F}_1=6\hat{i}+0\hat{j}$ kN, Team B pulls with force $\vec{F}_2=-4\hat{i}+4\hat{j}$ kN and Team C pulls with force $\vec{F}_3=-3\hat{i}-3\hat{j}$ kN.



(i) What is the magnitude of the force of Team A? — 1 mark

(ii) Which team will win the game? — 1 mark

(iii) In what direction is the ring getting pulled? — 2 marks

Answer key

Chapter 10: Vector Algebra, Practice Set B. Full step-wise solutions are in the solutions sheet.

Level 1

1	(D) $\dfrac{\hat{i}-\hat{j}+\hat{k}}{\sqrt{3}}$
2	(C) -19
3	(B) ± 1
4	(A) $3\sqrt{3}$
5	(C) $\sqrt{21}$
6	(C) A is true, but R is false
7	Area $= \dfrac{\sqrt{62}}{2}$ sq units

Level 2

8	d.c. $\dfrac{2}{7}, \dfrac{3}{7}, -\dfrac{6}{7}$; $\vec{r} = 4\hat{i}+6\hat{j}-12\hat{k}$, components $4, 6, -12$
9	$\pm(-2\hat{i}+4\hat{j}+4\hat{k})$
10	$\dfrac{\pi}{4}$
11	Area $= \dfrac{9}{2}$ sq units; perpendicular from A to $BC = 3$ units
12	$\lambda = 5$; projection vector $= \dfrac{8}{7}\hat{i}+\dfrac{24}{7}\hat{j}+\dfrac{12}{7}\hat{k}$
13	$\lambda = \dfrac{1}{2}$
14	$\overrightarrow{OC} = \dfrac{3\vec{a}-\vec{b}}{2}$

Level 3

15	$\vec{a}\times\vec{b} = \vec{b}\times\vec{c} = \vec{c}\times\vec{a}$; each has magnitude twice the area of the triangle formed by $\vec{a}, \vec{b}, \vec{c}$.
16	Vector area $= \dfrac{1}{2}[\vec{b}\times\vec{c}+\vec{c}\times\vec{a}+\vec{a}\times\vec{b}]$ (shown); collinear $\iff \vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}=\vec{0}$; $\hat{n} = \dfrac{\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}}{\lvert\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}\rvert}$
17	$6\hat{i}-6\hat{j}-3\hat{k}$
18	Area $= 4\sqrt{3}$ sq units
19	(i) proved; (ii) $\cos A = \dfrac{1}{3}$
20	(i) 6 kN; (ii) Team B; (iii) along $-\hat{i}+\hat{j}$, at $\dfrac{3\pi}{4}$ with the positive x-axis